

Arie Klootwijk's Sulfone-Containing Poly(hydroxy ether) Patent

A forensic technical review of SE 301 717 B

What the SO₂ bridge and near-equimolar chemistry were actually doing

SVERIGE	UTLÄGGNINGSSKRIFT nr 301 717
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SHELL INTERNATIONALE RESEARCH MAATSCHAPPIJ NV, HAAG, NEDERLÄNDERNA
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Sätt att framställa termoplaster av en dihydroxiförening och en diepoxiförening varvid
diepoxiföreningen eller båda innehåller en SO₂-grupp

Prepared for Tom Klootwijk

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Scientific-historical review - not a certified translation, legal opinion, regulatory submission or laboratory safety protocol.

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1 Executive finding

The most defensible conclusion is not that the patent concealed a different SO₂ technology. It is that it disclosed, unusually early, a reactive high-performance thermoplastic platform whose significance is easy to miss if it is read only as a fibre-and-moulding patent.

The diaryl-SO₂ unit is a neutral sulfone bridge in the polymer backbone, not sulfur-dioxide gas, a sulfonate, a biological signalling group or an ion-exchange site. The near-equipolar language is the ordinary but unforgiving mathematics of bifunctional step-growth polymerisation: a 1-4% imbalance can impose a hard ceiling on chain length. The genuinely forward-looking part is the integration of stoichiometry, monomer purity, dryness, solvent polarity, solubility and phase control to produce a *linear epoxy-derived polymer* that remains thermoplastic and carries a secondary hydroxyl at every ring-opening event.

The patent's preferred reaction - bisphenol S (BPS) plus its diglycidyl ether - produces what would now be recognised as a **sulfone-containing poly(hydroxy ether)**, related to phenoxy or high-molecular-weight epoxy resins. It is distinct from commercial fully aromatic polysulfone/polyethersulfone, because its chain contains hydroxypropyl segments and abundant pendant OH groups.

Those hydroxyls make the material more than an inert moulding resin. They provide adhesion, hydrogen bonding, blend compatibility and a route to later crosslinking or derivatisation. A later Shell patent explicitly used linear diphenol-diepoxy units and lightly crosslinked them through the resulting secondary hydroxyl groups. A later BPS high-molecular-weight epoxy patent applied closely related ratio, purity and molecular-weight logic to printed-circuit-board insulating films. These are strong signs of technological lineage, though not proof that a single commercial product ran continuously from the 1959 work.

1.1 Answer to the “what was he up to?” question

Arie Klootwijk and his co-inventors appear to have been doing four things at once:

1. **Turning epoxy chemistry into a linear engineering thermoplastic rather than a cured three-dimensional network.**
2. **Using the aryl-sulfone bridge to raise softening, heat-distortion and mechanical performance.**
3. **Solving a process-physics problem:** growing chains precipitated or crystallised too early, so the team selected polar, nonreactive solvents that kept the product mobile until high molecular weight was reached.
4. **Programming chain length and end groups** with near-equivalent bifunctional feeds, anhydrous phenolate catalyst and deliberate phenol-rich stopping.

That combination was genuinely sophisticated for a 1959 priority filing. It anticipated themes that are now familiar in high-performance polymers: reactive thermoplastics, thermoplastic-to-thermoset conversion, toughening of epoxy matrices, high-temperature dielectric films and structure-property tuning by polar aromatic groups.

1.2 What the evidence does *not* support

There is no documentary or chemical evidence that the patent was a disguised route to:

- controlled release or storage of SO₂;
- intrinsic proton or ion transport;
- biological sulfur signalling;
- drug delivery, tissue scaffolds or implants; or
- a hidden energetic or catalytic material unrelated to polymers.

Those hypotheses fail because the sulfur is locked in a stable neutral diaryl sulfone and every claim, example, property test and later citation stays within macromolecular materials technology.

1.3 Best modern application bets

The strongest opportunities, ranked by evidence, are:

- high-temperature dielectric, adhesive and build-up films for electronics;
- coatings, primers, hot-melt or reactive adhesives;
- thermoplastic tougheners and reactive interlayers in epoxy/composite systems;
- moulded, extruded or fibre products where a higher thermal window than BPA phenoxy is valuable;
- corrosion-resistant and barrier layers, subject to moisture and extractables testing; and
- membrane-blend or bio-device components as research hypotheses, not qualified uses.

Direct biological contact is a high-barrier proposition. Current hazard concern around BPS makes residual monomer, DGE-BPS, low oligomers and degradation products central. A high-molecular-weight polymer is not equivalent to free BPS, but that distinction must be demonstrated analytically and toxicologically.

1.4 The most revealing overlooked detail

Example 1 is often read as 0.97 mol BPS against 1.00 mol DGE-BPS. The patent says phenol present as salt is included in the ratio. Adding the 0.05 mol monosodium-BPS catalyst gives an effective BPS-derived feed of **1.02:1** - exactly the phenol-rich boundary of the preferred range. The catalyst is therefore also part of the functional-group bookkeeping. This is not an encoded secret, but it is a subtle, deliberate piece of molecular-weight and end-group design.

How this review separates evidence from imagination

The "hidden revolution" survives only where chemistry and later records support it.

RESEARCH HYPOTHESIS

osensor surfaces, affinity membranes, additive manufacturing
post-sulfonated ion conductors - each requires new evidence.

SUPPORTED EXTRAPOLATION

Reactive films, coatings, adhesives, PCB dielectric, epoxy toughener,
thermoplastic-to-thermoset conversion, selected membrane blends.

DIRECTLY SUPPORTED

Linear high-MW poly(hydroxy ether); sulfone raises thermal/impact metrics;
thermoplastic shaping; stoichiometry/solubility/dryness control.

Rejected: SO₂ release, secret biological signaling, intrinsic ion exchange, or implant claims.

2 Scope, method and confidence language

This review uses the user-supplied Swedish patent and its clean English translation as the primary basis. The British family publication and U.S. counterpart are used to resolve terminology and add directly corresponding comparison data. Later patents and literature are clearly labelled as context or extrapolation.

The report uses the following confidence language:

- **High:** directly stated/demonstrated in the original family or strongly validated by a later patent using substantially the same material logic.
- **Medium:** supported by close phenoxy/BPS analogues and by chemical mechanism, but not demonstrated for the exact Klootwijk polymer.
- **Low:** plausible only after substantial new chemistry or qualification.
- **Unsupported:** inconsistent with the claimed structure or lacking any documentary basis.

Historical numerical data are preserved in their published units; yield stress is also converted to MPa. No attempt is made to convert intrinsic viscosity into absolute molecular weight without a polymer-specific Mark-Houwink relation.

2.1 Key limitations

The patent provides single historical examples, not a modern replicated dataset. It lacks SEC/GPC, molecular-weight distribution, spectroscopic structure confirmation, branching analysis, statistical uncertainty, long-term creep/fatigue, dielectric testing, flame testing, extractables/leachables and biocompatibility. The modern application discussion is therefore an evidence-weighted research roadmap, not a product qualification.

3 What the patent literally covers

3.1 Identity, dates and inventors

SE 301 717 B names R. W. Kreps, Arie Klootwijk and J. M. Goppel as inventors and Shell Internationale Research Maatschappij N.V. as applicant. The priority is Netherlands application 240 201 of 15 June 1959. The Swedish application was filed on 15 June 1960 and the laid-open specification was published on 17 June 1968. The British family publication appeared in 1963, and the corresponding U.S. continuation was issued in January 1968.

The Swedish title can read as though the diepoxide or both compounds must contain SO_2 , but the body and Claim 1 are broader: the dihydroxy compound, the diepoxide, or both may contain the diaryl-sulfone bridge.

3.2 Claim 1 as a process-window claim

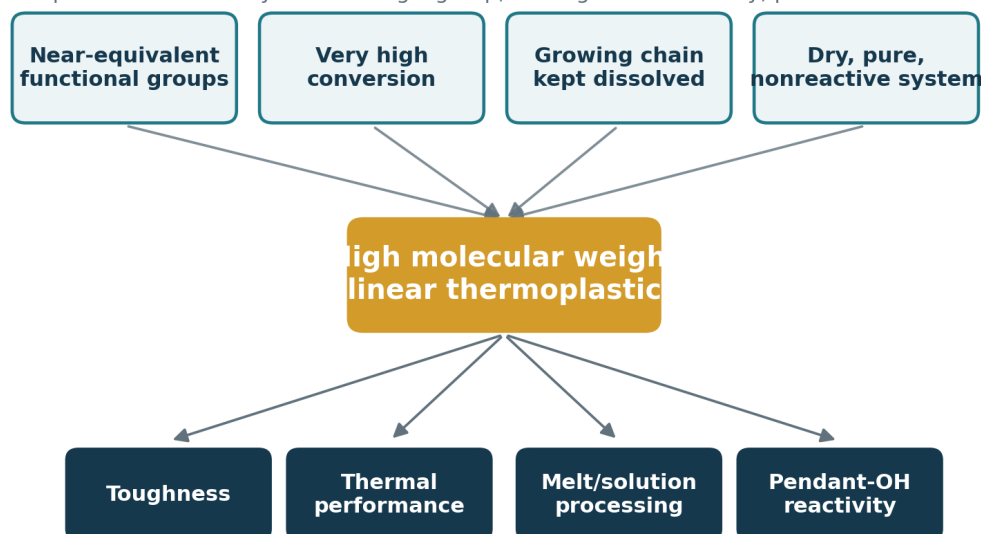
The core claim requires the combination of:

- one or more dihydroxy compounds;
- a substantially equimolar quantity of one or more diepoxy compounds;
- an aryl- SO_2 -aryl group in either or both reactant families;
- an alkaline catalyst;
- a substantially anhydrous ketone, nitrile, nitro compound, sulfoxide or sulfone solvent;
- solvent dielectric constant at least 5;
- solubility of both reactants and the growing polymer;
- low solvent reactivity toward phenol and epoxide; and
- continuation until intrinsic viscosity reaches at least 0.4.

This is not a claim to the sulfone group in isolation. It claims a coupled chemical and physical process that prevents premature termination.

The true inventive unit is a coupled process window

The patent does not rely on one magic group; it integrates chemistry, phase behavior and metrology.



3.3 Dependent-claim architecture

Claim	Theme	Practical meaning
2	Dihydric phenol geometry	Avoid ortho-related hydroxyl geometry when on the same ring.
3	Diepoxide class	Prefer di(epoxyalkyl) ethers, especially diglycidyl ethers.
4	Preferred pair	BPS plus DGE-BPS: sulfone in both feed components.
5	Higher-MW window	Keep product dissolved until intrinsic viscosity at least 0.8.
6	Higher solvent polarity	Prefer dielectric constant at least 10.
7	Anhydrous catalyst	Avoid adding water with the catalyst.
8-9	Catalyst identity	Use a salt of the participating dihydric phenol, especially a mono-alkali salt.
10	BPS purity proxy	BPS melting point at least 240 C.
11	DGE-BPS purity proxy	Diglycidyl ether melting point at least 162 C.

The claim ladder reveals the inventors' hierarchy: first secure the homogeneous high-conversion process, then narrow to geometry, molecular weight, polarity, dryness, catalyst integration and monomer purity.

3.4 The problem they said they solved

Earlier art allowed diphenylol sulfone and glycidyl chemistry but used ethanol or dioxane. The inventors attributed inferior products to either:

- crystallisation/precipitation of the product before sufficient molecular weight was reached; or
- solvent acting as a chain stopper.

Their solution was to keep the growing macromolecule in a solvent that dissolved it and did not attack either functional group. This is a phase-behaviour intervention as much as a reaction-chemistry intervention.

4 Chemical identity and mechanism

4.1 Bottom-line identification

The preferred material in SE 301 717 B is best described as a **sulfone-containing linear poly(hydroxy ether)**, or more specifically a **bisphenol-S poly(hydroxy ether)** when both preferred monomers are used. In later terminology it sits in the phenoxy-resin / high-molecular-weight epoxy family. It is not the same polymer as commercial polysulfone (PSU) or polyethersulfone (PESU), even though all contain diaryl-sulfone motifs.

The preferred feed pair is:

- **4,4'-sulfonyldiphenol**, historically called p,p'-diphenylol sulfone and now commonly called **bisphenol S (BPS)**; and
- its **diglycidyl ether (DGE-BPS)**, a bifunctional diepoxide.

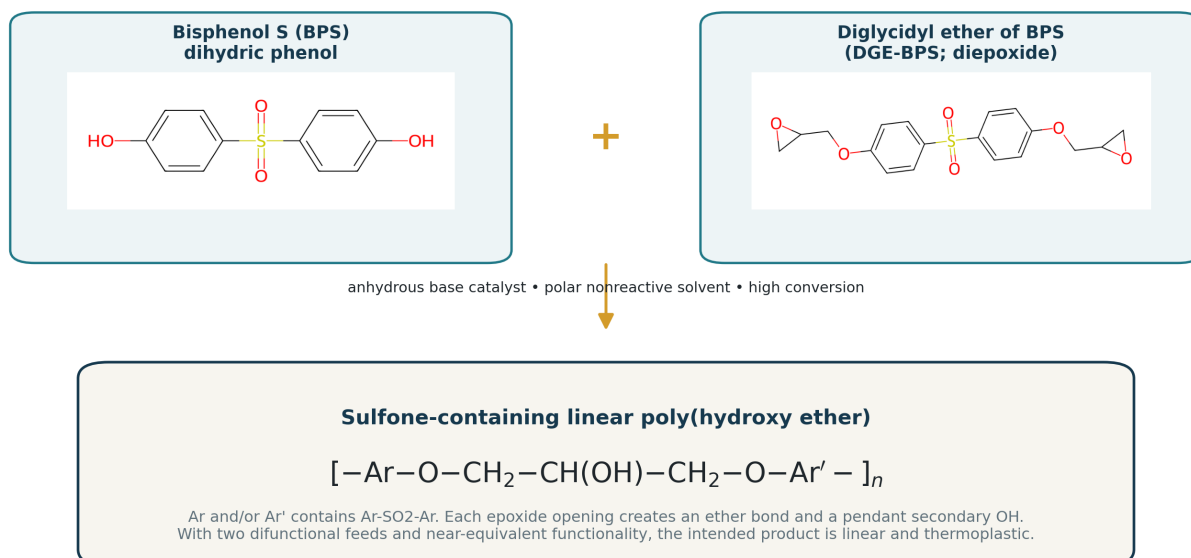
A base generates or transfers phenoxide. Phenoxide attacks an epoxide ring, normally at its less substituted carbon. Ring opening forms an aryl-ether bond and a secondary alkoxide, which is proton-transferred to a secondary alcohol while another phenoxide is generated. Repetition links two difunctional monomers into an almost linear chain.

A generic segment is:



where Ar and/or Ar' contains the neutral diaryl-sulfone bridge $\text{Ar}-\text{S}(=\text{O})_2-\text{Ar}$.

The chemistry actually claimed



4.2 Why this is an “epoxy” that remains thermoplastic

In conventional epoxy curing, epoxides react with multifunctional hardeners and form a three-dimensional network. Here, each principal monomer is nominally **difunctional**, and the patent aims for a nearly 1:1 balance of the two complementary functions. The ideal result is therefore a long, linear chain that softens and flows on heating rather than an infinite network.

Each epoxide opening creates a pendant secondary hydroxyl group. That has two consequences:

1. the polymer is polar, adhesive and capable of hydrogen bonding; and
2. it retains latent reactivity. Those hydroxyls can later be crosslinked or derivatized, and residual epoxide can react with them under stronger conditions.

The original patent tries to suppress unwanted side reactions and branching by choosing nonreactive solvents, excluding water and oxygen, controlling monomer purity, and stopping the reaction at the desired viscosity. A later Shell patent explicitly used the same family of linear diphenol-diepoxide units and lightly crosslinked them through the resulting secondary hydroxyl groups. That later development does not prove a concealed intention in 1959, but it validates the platform interpretation.

4.3 The subtle role of the monosodium BPS “catalyst”

The patent states that any dihydric phenol present as a salt must be counted as dihydroxy compound when calculating the molar ratio. This makes the examples more subtle than a casual reading suggests.

Example 1 charges:

- 1.00 mol DGE-BPS;
- 0.97 mol neutral BPS; and
- 0.05 mol monosodium BPS.

On the patent’s own accounting, the effective BPS-derived feed is $0.97 + 0.05 = 1.02$ mol per 1.00 mol diepoxide, at the phenol-rich edge of the preferred range. Example 6 uses the same effective ratio. Example 5 is essentially balanced at about 1.002:1 if its 0.002 mol sodium salt is included.

The salt is catalytic in the sense that phenoxide character can be transferred through proton exchange, but the BPS molecule carrying sodium is itself a difunctional species that can be incorporated. It therefore affects both kinetics and end-group/stoichiometric bookkeeping.

4.4 What can break linearity or cap chains

The ideal AA + BB model is only an approximation. Important deviations include:

- **water and alcohols**, which can open epoxides or create monofunctional ends;
- **hydrolyzable chloride and alpha-glycol impurities** in glycidyl ethers;
- **mono-functional phenols or epoxides**, including impurities;
- **o,p’-BPS isomer**, which changes geometry and may disrupt regularity, packing or effective functionality;
- **epoxide homopolymerization** or alkoxide attack on residual epoxide, which can branch;
- **secondary-hydroxyl/epoxide reaction**, especially with epoxide excess or harsher catalysis;
- **premature crystallization**, which physically removes growing chains from the reactive solution; and
- **thermal chain scission or rearrangement** during high-temperature pressing.

The patent’s solvent, dryness, purity and temperature requirements can therefore be read as one integrated strategy: keep both functional groups chemically available and keep the growing chains physically mobile until the desired molecular weight is reached.

4.5 Distinction from PSU and PESU

Commercial aromatic polysulfones are generally made by nucleophilic aromatic substitution between a bisphenolate and an activated aromatic dihalide such as 4,4’-dichlorodiphenyl sulfone. Their backbones are highly aromatic and contain aryl-ether and aryl-sulfone bonds, but no regularly spaced hydroxypropyl segments.

Klotwijk’s preferred polymer instead contains:

- aliphatic -CH₂-CH(OH)-CH₂- units created by epoxide opening;
- a secondary hydroxyl at every ring-opening event;
- potentially greater water affinity and chemical functionalizability; and
- a lower-temperature, more chemically reactive route than a fully aromatic S_NAr polymerization.

Calling the material simply “polysulfone” would therefore erase the feature that is most technologically interesting: it is a **reactive engineering thermoplastic** rather than only a heat-resistant fully aromatic thermoplastic.

5 Equimolarity: molecular-weight programming, not a cipher

5.1 Why the wording is central

The phrase “substantially equimolar” is not decorative patent language. In a linear step-growth reaction between a difunctional dihydroxy compound and a difunctional diepoxy, high chain length is possible only when:

- the two complementary functional groups are very close to equivalent;
- conversion is very high;
- water, alcohols and monofunctional impurities do not consume functions; and
- the growing polymer remains mobile enough for end groups to find each other.

The patent defines a dihydroxy:diepoxy molar ratio of about 1.04 to 0.98, preferably 1.02 to 0.99. Because each ideal molecule has two relevant functions, molecular ratio and functional-group equivalent ratio coincide.

5.2 Carothers-Flory relation

Let r be the ratio of the smaller initial number of functional groups to the larger, so $0 < r \leq 1$, and let p be the conversion of the limiting functions. The ideal number-average degree of polymerization is:

$$X_n = (1 + r) / (1 + r - 2rp)$$

At complete conversion of the limiting group:

$$X_{n,max} = (1 + r) / (1 - r)$$

The result is nonlinear. A seemingly small analytical or weighing error can impose a hard ceiling on chain length.

Dihydroxy:diepoxy feed	r used in equation	Ideal X_n ceiling at $p = 1$
1.040:1	0.9615	51
1.020:1	0.9804	101
1.010:1	0.9901	201
1.000:1	1.0000	unbounded in the ideal model
0.990:1	0.9900	199
0.980:1	0.9800	99
0.970:1	0.9700	65.7

Even exact stoichiometry is not enough. At $r = 1$, $X_n = 1/(1-p)$, so 99% conversion gives X_n about 100, 99.5% gives 200, and 99.9% gives 1,000. This is why the solvent and solubility claims are inseparable from the equimolar claim: the reaction must remain homogeneous long enough to drive the final fractions of a percent of conversion.

5.3 Re-reading Example 1

The easy but incomplete reading is 0.97 mol BPS versus 1.00 mol DGE-BPS. The patent expressly says to include BPS present as salt. Adding 0.05 mol monosodium BPS gives an effective BPS-derived total of 1.02 mol. That is exactly the upper edge of the preferred range.

Under the ideal model, a 1.02:1 feed gives an X_n ceiling of about 101 monomer molecules. In a symmetric BPS/DGE-BPS system, that is on the order of 50 alternating feed pairs. A rough mass estimate based on the two-monomer pair is in the tens of kilodaltons, consistent with the later industry's description of these materials as high-molecular-weight epoxy or phenoxy resins. This estimate is illustrative only: the patent reports intrinsic viscosity, not absolute molar mass, and branching, cyclic material and end-group chemistry alter the mapping.

5.4 Why a slight deliberate imbalance can be useful

A perfectly balanced polymer can retain two different reactive end groups and continue changing during hot processing. A controlled small excess can:

- cap both chain ends predominantly with the excess function;
- set a reproducible molecular-weight ceiling;
- reduce residual epoxide when phenol is in excess;
- stabilize the product against uncontrolled post-polymerization; and
- provide defined hydroxyl-rich end chemistry.

Example 4 makes this strategy explicit. After rapid polymerization at 150 C, about 2.5 g BPS (approximately 0.010 mol) is added to stop further growth. It is an end-group intervention, not merely a quench by cooling.

5.5 Why purity and dryness are stoichiometry controls

Every reactive impurity perturbs effective r or prevents p from approaching unity. Water and alcohol are especially damaging because they can be monofunctional chain terminators. A few mole percent of functionality loss is enough to cut the ideal degree of polymerization by an order of magnitude.

This is why the patent specifies:

- very low water content;
- anhydrous catalyst introduction;
- high-melting, purified p,p'-BPS;
- low o,p'-isomer content;
- low chloride in DGE-BPS; and
- a solvent that neither reacts with phenol nor epoxide.

A later high-molecular-weight BPS epoxy patent repeats the same design logic using an epoxy:phenolic-OH equivalent ratio of 1:0.94-1.06, high 4,4'-BPS purity and low hydrolyzable chlorine. A conventional polysulfone patent likewise states that a deviation as small as 5% from equal molar amounts seriously reduces molecular weight. These later documents independently confirm that the original ratio language describes ordinary but critical polymer physics, not coded chemistry.

6 What the SO₂ bridge does - and does not do

6.1 It is a diaryl sulfone, not "SO₂ gas in a polymer"

The structural element is Ar-S(=O)₂-Ar: sulfur in a high oxidation state, doubly bonded to two oxygens and covalently bound to two aromatic rings. It is:

- neutral rather than ionic;
- a backbone bridge rather than a pendant acid;
- strongly polar;
- relatively rigid and bulky; and
- electronically withdrawing from the aromatic rings.

It is not SO₂ dissolved in the material, not a sulfite, not a sulfonate and not a sulfonyl chloride. Under intended use it is not expected to liberate sulfur dioxide as a controlled function. Any sulfur-containing products of destructive pyrolysis would be decomposition products, not the claimed technology.

6.2 Structure-property role

The most defensible molecular effects are:

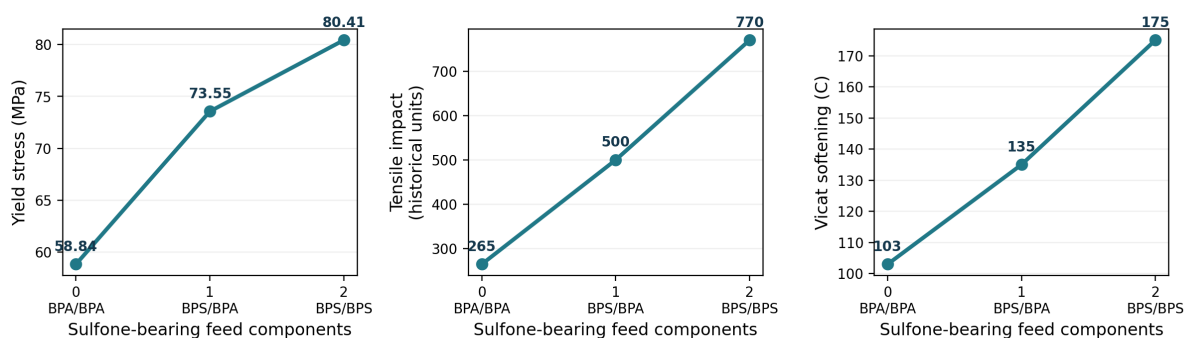
- **higher cohesive energy and dipole-dipole interaction**, raising resistance to segmental motion;
- **greater backbone rigidity** than an isopropylidene or simple alkylene bridge;
- **higher softening/heat-distortion response**, directly shown in the U.S. comparison;
- **altered crystallisation and packing**, seen in the crystalline all-sulfone product;
- **enhanced polarity and solvent interactions**; and
- **increased phenol acidity**, which makes phenolate generation easier, while the electron-withdrawing ring can also moderate nucleophilicity.

The original U.S. counterpart makes the comparison unusually explicit. In its first three polymers, sulfone-bearing feed components are increased from zero to one to two:

Polymer	Feed pair	Intrinsic viscosity	Yield stress	Tensile impact*	Vicat softening
1	BPA + DGEBA	0.57	600 kgf/cm ² (58.8 MPa)	265	103 C
2	BPS + DGEBA	0.67	750 kgf/cm ² (73.5 MPa)	500	135 C
3	BPS + DGE-BPS	0.70	820 kgf/cm ² (80.4 MPa)	770	175 C

*Historical patent units are retained; the method is not mapped to a modern SI impact standard.

U.S. counterpart: the first three comparison polymers show a strong sulfone-content trend



Intrinsic viscosity also rises (0.57 → 0.67 → 0.70), so a modern matched-molecular-weight series is still needed.

The trend is compelling, but not perfectly isolated: intrinsic viscosity also rises across the series. A modern experiment should compare matched absolute molecular weights and identical thermal histories.

6.3 Polarity is beneficial and costly

The sulfone plus regularly spaced secondary OH groups gives strong polar interactions and potential adhesion. It also increases water affinity. In the Swedish all-sulfone Example 1, mass gain after 14 days at 20 C reaches 5.6 wt% in water, 5.1 wt% in 5% NaOH and 5.4 wt% in 5% sulfuric acid. The patent says no chemical change occurred, but modern use would require dimensional, mechanical, dielectric and extractables retention after conditioning.

6.4 Why it is not intrinsically ion-conducting

An ion-exchange polymer needs fixed ionic groups such as sulfonate, phosphonate or quaternary ammonium. A neutral diaryl sulfone provides dipolar character but no dissociable charge. Post-sulfonation could create an ionomer, but that would be a new material with different water uptake, morphology and hydrolytic behaviour.

7 Experimental evidence in the Swedish examples

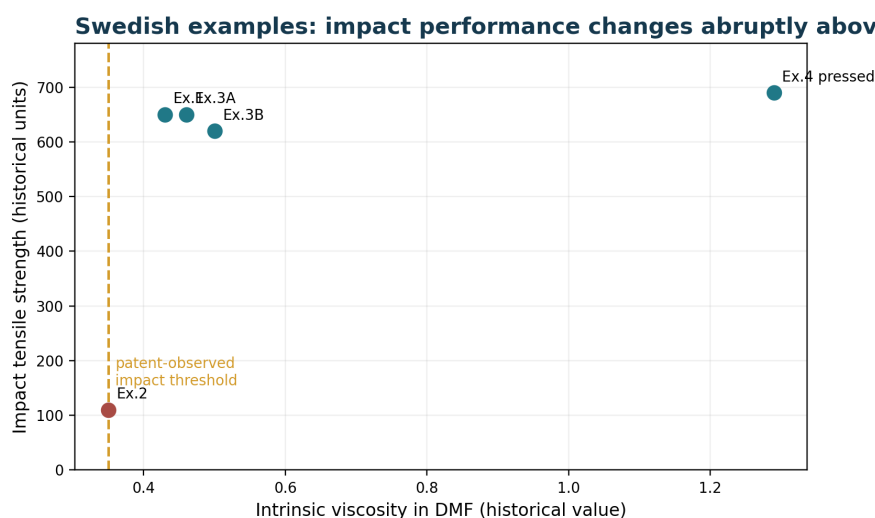
7.1 Example map

Example	Core architecture	Process	IV in DMF	Selected outcome
1	DGE-BPS + BPS	acetophenone, 100 C, 60 h	0.43	yield 75.5 MPa; impact 650; HDT 154 C; crystalline melt about 230 C
2	DGE-BPS + BPS	same, 20 h	0.35	impact only 110 despite similar yield
3	DGE-BPS + BPS	reheated to 150 C to redissolve	0.46 / 0.50	impact 650 / 620
4	DGE-BPS + BPS	150 C, 50 min; BPS stop	1.37, then 1.29 after pressing	impact 690; very high IV reached rapidly
5	DGE-BPS + BPS	sulfolane, 105 C, 24 h	0.42	alternative polar sulfone solvent works
6	DGEBA + BPS	Example-4 type route	0.82	HDT 114 C; Tg about 130 C; water uptake 1.8%

Full transcribed data are supplied in 04_Data/patent_examples.csv.

7.2 Molecular-weight threshold and toughness

The patent's clearest causal result is the contrast between Example 2 and Examples 1/3. At IV 0.35, tensile impact is 110 historical units. At IV 0.43-0.50, it is roughly 620-650. Yield stress changes much less. The inventors concluded that good impact strength required intrinsic viscosity above 0.35.



This is exactly what a linear thermoplastic would be expected to show: once chains are long enough for entanglement and effective stress transfer, toughness rises nonlinearly. The data are sparse and unreplicated, but the qualitative threshold is strong.

7.3 The crystallisation paradox

Crystallisation appears in two roles:

1. **A reaction failure mode:** if polymer crystallises out too early, active chain ends lose mobility and molecular weight stalls.
2. **A final material feature:** Example 1 is crystalline and birefringent, with a sharp diffraction pattern and crystallite melting near 230 C.

The invention therefore does not eliminate crystallinity. It delays phase separation until chain growth is sufficient. Example 3 makes the concept operational: temperature is raised to 150 C specifically to redissolve the crystallising product and continue polymerisation.

7.4 Thermal aging at 160 C

Example 1 yield stress rises from 770 to 1,000 kgf/cm² after 24 hours at 160 C and stays near that level through 100 hours. Possible explanations include annealing, increased crystallinity, physical aging or post-reaction. The result cannot be called unambiguously beneficial because post-aging elongation and impact were not reported. A stiffer/stronger but more brittle material could produce the same yield-stress trend.

7.5 Melt-processing warning at 290 C

Example 4 reaches IV 1.37 but falls to 1.29 after pressing at 290 C, about a 5.8% decrease. Measurement scatter is unknown, so chain scission cannot be proven. Nevertheless, it signals that a modern process window must map molecular-weight retention versus temperature and residence time.

7.6 Architecture tuning in Example 6

Replacing DGE-BPS with DGEBA leaves sulfone only in BPS. The product has T_g about 130 C and HDT 114 C, both below the all-sulfone result, but water uptake is only 1.8% after 140 hours. The comparison shows that sulfone content and hydroxyether density can tune heat/water trade-offs rather than produce one fixed material.

8 Was the ordinary application language camouflage?

8.1 Why patents sound broader and more mundane than the chemistry

Patents commonly list familiar forms - fibres, films, mouldings, castings, extrusions - to demonstrate utility and obtain broad practical coverage. Such wording can make the underlying materials concept seem less ambitious. Here, however, the claims themselves are technically specific and transparent. They focus on:

- equivalent balance;
- solvent polarity and nonreactivity;
- homogeneous chain growth;
- dryness;
- phenolate catalyst;
- molecular-weight threshold; and
- monomer purity.

That is not the pattern of a document hiding a different mechanism. It is the pattern of a process patent written to protect a difficult polymerisation window.

8.2 The genuinely revolutionary layer

The forward-looking idea is **process now, react later**:

- Build a high-molecular-weight linear chain with epoxy chemistry.
- Retain a secondary OH at every ring opening.
- Shape it as a thermoplastic.
- Optionally exploit the OH functionality later for adhesion, crosslinking or interfacial coupling.

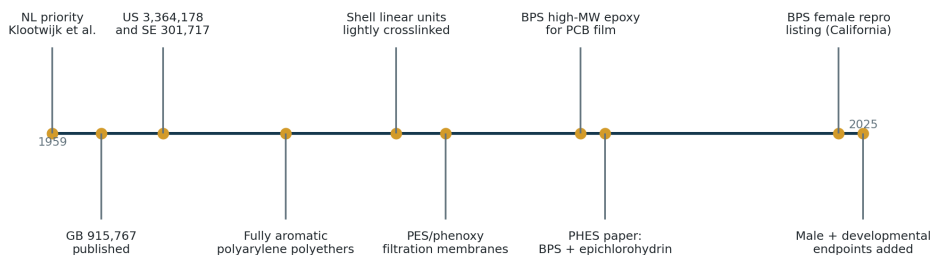
The original patent does not fully develop the fourth step, but the chemistry makes it possible. The later Shell thermoplastic/thermoset patent makes this latent capability explicit. Thus the “hidden” element is better described as an **under-emphasised platform property**, not an obfuscated alternate invention.

8.3 Comparison with a conventional epoxy thermoset

Feature	Klootwijk linear poly(hydroxy ether)	Conventional multifunctional epoxy cure
Principal functionality	two + two	often more than two effective network functions
Target topology	long, almost-linear chain	3D network
Processing	solution or melt thermoplastic	liquid/low-MW resin then irreversible cure
Reworkability	potentially melt/re-dissolve before crosslinking	generally not melt-reprocessable
OH generation	one secondary OH per epoxide opening	also generated, but embedded in network
Latent reactivity	high; can be post-crosslinked	cure consumes the process window
Key risk	stoichiometric imbalance and chain stopping	gelation, cure shrinkage, exotherm

9 Historical and technological lineage

A technological lineage - not proof of one continuous product, but repeated validation of the design logic



9.1 1959-1968: the Klootwijk family of disclosures

The Netherlands priority was filed in 1959. GB 915,767 A published in 1963 and gives an English-language family description. SE 301 717 B and US 3,364,178 A were published in 1968. The U.S. document is a continuation and focuses more explicitly on the thermoplastic poly(hydroxyalkyl diphenyl sulfone) product and the sulfone-content comparison.

9.2 Relation to fully aromatic polysulfone/polyethersulfone

A later dominant route to high-performance sulfone thermoplastics uses a bisphenolate plus activated diaryl dihalide in a nucleophilic aromatic substitution. US 4,108,837 describes high-molecular-weight polyarylene polyethers made in sulfoxide/sulfone solvents under dry, near-equimolar conditions. It states that even a 5% variation from equal molar amounts seriously reduces molecular weight and that polymer remaining dissolved gives higher molecular weight.

That later route independently confirms the same process logic but yields a different backbone: fully aromatic aryl ether-sulfone without the hydroxypropyl units. It helps explain why the Klootwijk material should not simply be called PSU or PESU.

9.3 1987: latent OH reactivity becomes explicit

EP 0 249 262 A2 describes linear diphenolic/diepoxide units that are lightly crosslinked through the secondary hydroxyls generated during formation. Its goal is a thermoplastic composition with thermosetting processing characteristics and a modulus/Tg/toughness balance. The Klootwijk family is recorded among the cited prior art.

This is the strongest evidence that the original linear poly(hydroxy ether) was a platform capable of being pushed toward thermoplastic-thermoset hybrids.

9.4 1991: phenoxy in filtration membranes

US 5,076,935 uses a phenoxy resin blended with polyethersulfone in porous filtration membranes. It describes uses including sterile vent filtration for fermentation tanks and reports improved membrane strength and easier melt sealing. The exact BPS poly(hydroxy ether) is not the disclosed phenoxy, so this is a close-analogue application, not direct proof.

9.5 2002: BPS high-molecular-weight epoxy for PCB film

US 6,451,878 B1 claims high-molecular-weight epoxy resin made from a difunctional epoxy and a dihydric phenol containing 70-100% BPS. It specifies:

- weight-average molecular weight 10,000-200,000;
- epoxy:phenolic-OH equivalent ratio 1:0.94-1.06;
- high 4,4'-BPS purity;
- low hydrolyzable chlorine/alpha-glycol impurity;

- low catalyst-derived ionic contamination; and
- use in printed-circuit-board insulating film.

It reports T_g up to about 155 C and thermal expansion around 50-60 ppm/C versus about 70 ppm/C for conventional epoxy. The echoes of the 1959 logic - equivalent balance, BPS purity, glycidyl impurity, high molecular weight and film processing - are unmistakable.

9.6 2004: material-class confirmation

Lü, Zheng and Tian published a material explicitly named **poly(hydroxyether sulfone)** synthesised from BPS and epichlorohydrin. They characterised hydrogen-bonding interactions involving OH, sulfone and ether sites. Their route and research question differ from the patent, but the paper confirms that the material identity inferred here is not a retrospective invention.

9.7 Why the Klootwijk route may not have become the dominant “polysulfone”

The historical record in this package does not establish commercial decisions. Chemically plausible reasons include:

- demanding purity and equivalent control;
- solvent-heavy polymerisation and isolation;
- strong crystallisation/phase sensitivity;
- pendant-OH water uptake;
- a narrower melt-stability window than fully aromatic PSU/PESU;
- high-melting DGE-BPS monomer; and
- later availability of fully aromatic sulfone polymers with exceptional continuous-use performance.

These are inferences to test, not documented reasons for abandonment.

10 Application analysis

10.1 Evidence tiers

The applications below are separated into three evidence tiers.

1. **Directly supported:** stated or demonstrated in the Klootwijk patent or its U.S. counterpart.
2. **Technologically supported extrapolation:** supported by later patents or established use of closely related phenoxy / BPS high-molecular-weight epoxy chemistry.
3. **Research hypothesis:** chemically plausible, but the exact Klootwijk polymer has not been validated for the use.

10.2 Highest-confidence uses

10.2.1 High-temperature films, molded parts, fibres and extrusions

The original patent directly identifies threads, fibres, casting, extrusion, injection molding and rolling. Its examples show hot pressing, high yield stress, substantial impact performance above a molecular-weight threshold, a heat-distortion temperature of 154 C for the all-sulfone example, and a crystalline melt near 230 C. These data support a genuine engineering-thermoplastic objective.

10.2.2 Coatings, primers, laminating resins and adhesives

The repeat chain contains abundant secondary hydroxyl groups and aromatic ether/sulfone segments. Closely related linear poly(hydroxy ethers), commonly called phenoxy resins, are used for coatings, laminates, primers, hot-melt adhesives, wire varnishes and resin modification. Hydroxyl groups provide substrate interaction and can be crosslinked with isocyanate, melamine-formaldehyde, phenolic or related chemistry. The sulfone version should be screened where a higher glass transition or heat resistance is valuable.

10.2.3 Electrical and printed-circuit dielectric films

A later patent on high-molecular-weight BPS epoxy resins uses essentially the same difunctional-epoxy plus BPS design space for printed-circuit-board build-up insulating film. It reports molecular weights of 10,000-200,000, BPS-rich compositions, a T_g up to about 155 C and lower thermal expansion than conventional high-molecular-weight epoxy. This is the strongest concrete modern validation of the platform.

10.2.4 Toughener, rheology modifier and reactive interlayer for epoxy composites

A high-molecular-weight linear epoxy/phenoxy phase can increase film strength, impact resistance, flexibility and adhesion while retaining compatibility with epoxy formulations. The pendant hydroxyls enable coupling to thermoset matrices. A later Shell patent deliberately converts linear diphenol-diepoxy units into lightly crosslinked thermoplastic/thermoset hybrids through the secondary hydroxyls. For carbon-fibre laminates, copper-clad laminates, structural adhesives and protective coatings, this is a highly plausible continuation of the original idea.

10.3 Medium-confidence uses

10.3.1 Barrier and multilayer films

Phenoxy films are known for oxygen-barrier and heat-seal behavior, while the Klotwijk material adds a rigid polar sulfone. The combination could be useful as a thin functional layer or tie layer. The patent does not report gas transmission, moisture-vapor transmission, taste/odor, migration or long-term hydrolysis. Food-contact or biomedical packaging would require exceptionally rigorous BPS and oligomer extractables work.

10.3.2 Corrosion-resistant and chemically resistant coatings

The patent reports immersion in water, 5% sodium hydroxide and 5% sulfuric acid without an observed chemical change, although liquid uptake reached roughly 5 wt%. Related phenoxy resins are strong coating binders. A modern program should measure adhesion and mechanical retention after exposure rather than relying on visual chemical stability.

10.3.3 Membrane blend modifier

A later patent demonstrates filtration membranes made from blends of polyethersulfone and a phenoxy resin, including sterile vent filtration for fermentation tanks. The exact Klotwijk BPS poly(hydroxy ether) was not tested there. Its higher polarity, hydroxyl density, solvent compatibility and thermal sealability make it a credible blend candidate, but pore formation, fouling, extractables and sterilization stability would have to be measured.

10.3.4 Reactive additive-manufacturing feedstock

A thermoplastic that can be shaped first and crosslinked second is conceptually attractive for printed electronics, hot-melt deposition or composite additive manufacturing. The original polymer's 230-290 C processing range and the observed intrinsic-viscosity loss during 290 C pressing make melt stability a decisive unknown.

10.4 Bioengineering-facing hypotheses

10.4.1 Functionalizable biosensor or microfluidic surfaces

Secondary hydroxyls offer handles for surface activation or grafting. A sulfone-rich aromatic backbone may offer dimensional and thermal stability. However, the patent contains no protein-adsorption, sterilization, cytotoxicity or extractables data. Use should begin with non-contact device layers or external components, not implantation.

10.4.2 Bioprocess and sterile-filtration components

The material could be investigated as a toughening or heat-sealing component in a PES membrane blend, or as a robust housing/adhesive layer. Related phenoxy/PES membrane art supports the concept, but the exact polymer's water uptake and possible BPS residuals are major design constraints.

10.4.3 Chromatography support or affinity membrane

Hydroxyl-rich polymers can be derivatized, and phase-inversion processing might create porous structures. This is a research hypothesis only. Neutral hydroxyl and sulfone groups do not themselves create ion-exchange capacity.

10.5 Low-confidence or unsupported interpretations

10.5.1 Proton-exchange or ion-exchange membrane

The diaryl SO₂ bridge is neutral. It is not a sulfonic acid or sulfonate. Ion conductivity would require a separate chemical modification such as post-sulfonation, which creates a materially different polymer and may compromise hydrolytic stability.

10.5.2 SO₂ gas storage, release or biological sulfur-dioxide signaling

Nothing in the patent supports controlled SO₂ release. The sulfur is covalently locked in a diaryl sulfone. Thermal decomposition chemistry is not a functional release mechanism and was not claimed or tested.

10.5.3 Implant, scaffold or drug-delivery material

There is no evidence of biocompatibility, biodegradation, sterilization, extractables or chronic exposure safety. Modern concern about residual bisphenol S makes direct biological-contact use a high-barrier proposition until proven otherwise.

10.6 Practical ranking

Application	Evidence	Confidence	Main unresolved issue
PCB dielectric/adhesive film	later same-chemistry patent	High	dielectric loss, ionic contamination, modern processability
Coating/primer/adhesive	phenoxy class + pendant OH	High	solvent system, cure window, residuals
Epoxy/composite toughener	mechanism + later Shell hybrid	High	phase behavior and cure kinetics
Molded/extruded high-temperature part	original patent	High	long-term creep and oxidative aging
Fibre/filament	original patent	High	spinning window and hydrothermal stability
Corrosion-resistant coating	class evidence + immersion	Medium-high	property retention after exposure
Barrier/tie-layer film	phenoxy class	Medium	migration and moisture sensitivity
PES membrane blend modifier	related membrane patent	Medium	pore morphology, fouling, leachables
Printed/reactive thermoplastic	platform logic	Medium-low	230-290 C melt stability
Biosensor/microfluidic surface	functionalizable OH	Medium-low	sterilization, adsorption, extractables
Ion-exchange membrane	only after new chemistry	Low	neutral backbone, requires sulfonation
Implant/scaffold	no direct support	Low	biocompatibility and BPS residuals
SO ₂ release/storage	chemically mismatched	Unsupported	SO ₂ is a stable covalent bridge

11 A bioengineering reading

11.1 What is genuinely interesting to a biological engineer

The repeat structure combines four useful motifs:

1. a rigid aromatic skeleton;
2. polar, thermally robust sulfone groups;
3. flexible hydroxypropyl-ether links; and
4. a high density of secondary hydroxyls.

That combination suggests a functional surface or blend component rather than an intrinsically bioactive polymer. The OH groups can support covalent derivatisation, hydrogen-bonding and adhesion. The sulfone/aromatic portion can provide stiffness, heat resistance and compatibility with other high-performance polymers.

11.2 Plausible research directions

11.2.1 Non-contact biosensor and microfluidic device layers

A BPS poly(hydroxy ether) could be screened as an adhesive dielectric, protective film or functionalised substrate in a device where the material is separated from the sample by another barrier. The most immediate value would be mechanical/electrical and interfacial, not biological.

11.2.2 Functionalised membranes

The OH groups could be activated or grafted, and the polymer may blend with PES or related membrane polymers. Research questions include:

- whether the BPS version increases or decreases water wettability relative to BPA phenoxy;
- how it changes phase inversion and pore morphology;
- whether OH/sulfone interactions reduce or increase nonspecific protein adsorption;
- whether it improves toughness and heat sealing; and
- whether low-molecular-weight extractables can be driven to acceptable levels.

11.2.3 Bioprocess hardware and sterile filtration

The patent shows thermal exposure at 160 C and a high HDT in one composition, while related phenoxy/PES art demonstrates sterile vent membranes. This supports testing in external housings, seals, membrane supports or adhesive layers. It does not establish steam-autoclave, gamma, EtO or repeated clean-in-place stability.

11.2.4 Reactive chromatography or affinity support

The pendant OH groups are potential attachment points after chemical activation. However, the polymer is not intrinsically an ion exchanger, and the aromatic BPS content increases the burden of extractables and nonspecific adsorption characterisation.

11.3 Why direct biomedical use is not the default

The original work predates current biological-evaluation and chemical-characterisation expectations. It does not address:

- residual BPS or DGE-BPS;
- low oligomers;
- hydrolyzable chloride;
- catalyst metals;
- degradation during sterilisation;
- cell response;
- sensitisation/irritation;
- systemic exposure; or
- long-term implantation.

Direct implant or drug-contact proposals should therefore be treated as low-confidence until a complete chemical and biological risk programme is passed.

12 BPS, polymer safety and extractables

12.1 Monomer hazard is not automatically polymer hazard

A purified high-molecular-weight polymer has very different absorption, mobility and biological availability from free BPS. It would be scientifically incorrect to equate the repeat unit with exposure to the monomer. It would be equally incorrect to ignore:

- unreacted BPS;
- DGE-BPS;
- cyclic or low oligomers;
- hydrolysis or thermal-degradation products; and
- surface-enriched low-molecular-weight fractions.

The risk question is therefore **what can migrate or be generated under actual use**, not merely what repeat unit is drawn.

12.2 Current BPS hazard context

California OEHHA lists BPS for female reproductive toxicity (effective 2023), male reproductive toxicity (effective 2025) and developmental toxicity (effective December 2025). EFSA has also reviewed BPS studies and identified the need for further data. These listings do not determine the safety of every BPS-containing polymer, but they materially raise the evidentiary standard for food-contact, medical and chronic-exposure applications.

12.3 Minimum analytical package for sensitive uses

A modern programme should include:

- targeted LC-MS/MS for BPS and DGE-BPS;
- non-target high-resolution MS for oligomers and transformation products;
- headspace/GC-MS for residual solvents and volatile degradation products;
- ICP-MS for sodium and other catalyst residues;
- total and hydrolyzable chlorine;
- extraction under exaggerated and simulated-use conditions;
- mass balance and surface-area-normalised reporting;
- sterilisation and hydrothermal aging before repeat extraction; and
- molecular-weight and spectroscopy checks after aging.

12.4 Processing safety

The patent examples use hot acetophenone or sulfolane, alkaline phenolate, epoxide monomers, nitrogen blanketing and in one case a sealed tube. Reproduction requires local hazard evaluation, reaction calorimetry, pressure relief, solvent exposure control and validated oxygen-exclusion practice. The historical recipe is evidence, not a modern safe-operating procedure.

13 Critical evidence audit

13.1 What the patent proves reasonably well

- A BPS/DGE-BPS system can be driven to a high enough intrinsic viscosity to form a strong, melt-processable thermoplastic.
- Keeping the polymer dissolved until greater molecular weight is reached materially improves impact behavior.

- Sulfone-containing monomer choices give higher softening and thermal performance than the BPA-only comparison in the U.S. counterpart.
- The material can be pressed into plates and can survive short-term exposure to 160 C without loss of yield stress.
- Strongly acidic and alkaline aqueous media are absorbed but did not cause an obvious chemical change in the historical test.
- The process is sensitive to solvent, water, purity, isomer composition, chloride and the reaction's physical state.

13.2 What it does not prove

The patent is not a modern materials qualification package. It provides no:

- absolute molecular weights or molecular-weight distributions by SEC/GPC;
- end-group quantification or branching analysis;
- NMR, FTIR or elemental confirmation of repeat structure;
- statistically replicated mechanical data or uncertainty estimates;
- creep, fatigue, fracture-toughness or environmental stress-cracking data;
- dielectric constant, dielectric loss, breakdown strength or ionic-conductivity data;
- oxygen index, UL-94 or smoke/toxic-gas results;
- oxygen- or water-vapor-transmission data;
- hydrolytic aging, autoclave, gamma, electron-beam or EtO sterilization testing;
- extractables/leachables or residual monomer/oligomer data;
- cytotoxicity, sensitization, irritation or implantation data; or
- direct commercial application evidence.

13.3 Important alternative readings of the reported data

13.3.1 Yield stress after 160 C aging

Yield stress rises from 770 to 1,000 kgf/cm² after 24 hours and remains there through 100 hours. This could reflect useful annealing or crystallization, but higher yield stress alone can also accompany reduced ductility or embrittlement. Impact and elongation after aging were not reported.

13.3.2 "Chemical resistance" versus liquid uptake

After 14 days, mass gain is about 5.1-5.6 wt% in water, 5% NaOH and 5% sulfuric acid. The statement that no chemical change occurred is encouraging but does not establish dimensional stability, retained mechanical properties, molecular-weight retention or absence of extractables.

13.3.3 Intrinsic viscosity loss during pressing

Example 4 falls from intrinsic viscosity 1.37 to 1.29 after pressing at 290 C, a decrease of about 5.8%. Without replicate measurements, the difference cannot be assigned definitively to chain scission. It is nevertheless a clear reason to map residence-time and temperature stability before modern melt processing.

13.3.4 Crystallization as both failure mode and property

The invention prevents **premature** crystallization during chain growth. The final all-sulfone product is reported as crystalline, birefringent and melting near 230 C. Thus crystallinity is not rejected; its timing is controlled. Modern work should separate reaction-induced precipitation, equilibrium crystallinity, annealing and degradation.

13.3.5 Sulfone comparison is compelling but not perfectly isolated

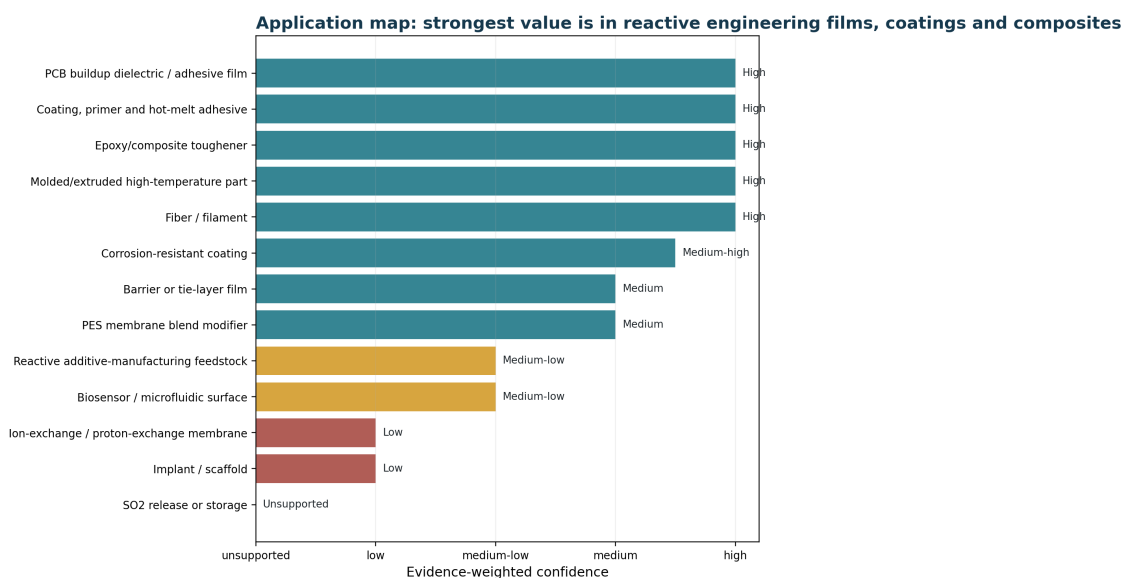
In the U.S. Table 1, the first three polymers have intrinsic viscosities 0.57, 0.67 and 0.70 as sulfone content increases. The corresponding rise in yield, impact and Vicat is consistent with a real sulfone effect, and the patent states that interpretation. Still, molecular weight also changes, so a modern controlled series should match molecular weight and thermal history before attributing the full effect to structure.

13.4 The “hidden revolution” hypothesis, tested

Hypothesis	Evidence for	Evidence against / limitation	Finding
The SO ₂ motif was a coded gas-release or biological-sulfur concept	none in claims, examples or family	covalent neutral diaryl sulfone; all work concerns polymer properties	Rejected
Equimolarity concealed a special self-assembly principle	narrow ratio, striking property threshold	exactly predicted by step-growth stoichiometry; repeated in later patents	Rejected as concealment; accepted as critical molecular-weight control
The ordinary uses hide a broader reactive platform	pendant OH, linear epoxy thermoplastic, later crosslinking patent	original never claims biological or dynamic functions	Partly supported: platform broader than the listed moulding/fibre examples
The work anticipated high-performance sulfone engineering plastics	early priority, thermal/impact data, sulfone comparison	chemistry is not the later fully aromatic PSU/PESU route	Supported, with a distinct material class
It anticipated electronics-grade BPS phenoxy/epoxy films	later BPS high-MW epoxy PCB patent, similar purity/ratio logic	no proof of direct commercial continuity	Strong technological lineage, not proven product lineage

13.5 Overall evidentiary conclusion

The document is not an obfuscated patent for a different field. Its real advance is visible in plain chemistry but easy to underestimate: it treats solubility, phase behavior, water, purity, equivalent balance and end-group chemistry as a single high-molecular-weight process window. The later industrial record supports that interpretation.



14 Modern reproduction and validation program

14.1 Objective

Recreate the three most informative architectures, quantify the relationship between stoichiometry, conversion, topology and properties, and decide whether the BPS poly(hydroxy ether) is most promising as a free thermoplastic, a reactive film/adhesive, a thermoset toughener or a membrane blend component.

This is a research plan, not a turnkey operating procedure. A competent laboratory should perform a formal hazard review for epoxides, strong bases, hot polar solvents, pressure and oxygen exclusion before execution.

14.2 Phase 1 - identity, purity and analytical standards

14.2.1 Monomers

Characterize BPS and DGE-BPS before polymerization:

- BPS 4,4'/2,4' isomer ratio by HPLC or quantitative NMR;
- water by Karl Fischer titration;
- DSC melting profile and impurity shoulders;
- DGE-BPS epoxy equivalent;
- hydrolyzable and total chlorine;
- alpha-glycol and other ring-opened impurity content;
- LC-MS profile of low-level monofunctional or oligomeric species; and
- metals by ICP-MS when electrical or biological uses are contemplated.

Prepare analytical standards for BPS, DGE-BPS and representative low oligomers so that later extractables work is quantitative rather than merely qualitative.

14.2.2 Benchmark architecture set

Use a 2 x 2 monomer matrix:

1. BPA + DGEBA: no sulfone control;
2. BPS + DGEBA: sulfone in the diphenol only;
3. BPA + DGE-BPS: sulfone in the diepoxide only; and
4. BPS + DGE-BPS: sulfone in both monomers.

This mirrors the logic of the U.S. comparison and separates sulfone dose from molecular weight.

14.3 Phase 2 - stoichiometry and process-window design

14.3.1 Effective ratios

Run effective dihydroxy:diepoxy ratios of approximately 0.98, 0.99, 1.00, 1.01 and 1.02, explicitly counting any phenol present as catalyst salt. Include a controlled monofunctional end-capper series.

14.3.2 Catalyst series

Start with anhydrous monosodium BPS to reproduce the patent. Then compare low-residue alternatives appropriate to the intended application. For electronic use, catalyst-derived Na, N and P must be tracked. For biological use, all catalytic residues and counterions must be included in the extractables assessment.

14.3.3 Solvent and phase-state series

Reproduce acetophenone and sulfolane conditions, then evaluate a smaller modern solvent set selected by:

- ability to dissolve both monomers and the growing polymer;
- chemical inertness toward epoxide and phenol;
- boiling point and pressure requirements;
- dielectric behavior at reaction temperature;
- ease of polymer isolation and solvent removal; and
- worker and environmental risk.

Measure turbidity or light scattering during reaction. The central patent claim can be tested directly by correlating the onset of phase separation with molecular weight and final toughness.

14.3.4 In-process monitoring

Use complementary measurements:

- ATR-FTIR or Raman for epoxide conversion;
- reaction calorimetry for heat release and rate changes;
- online or frequent-sample rheology/solution viscosity;
- SEC/GPC with multi-angle light scattering where solubility allows;
- quantitative NMR for end groups at lower molecular weight; and
- mass balance on isolated polymer and soluble oligomer fractions.

14.4 Phase 3 - molecular architecture

For each material determine:

- Mn, Mw and dispersity;
- intrinsic viscosity in the historical DMF method for bridge-to-patent comparability;
- branching index using SEC-viscometry or light-scattering methods;
- residual epoxide and phenolic/secondary OH content;
- soluble gel fraction and evidence of microgel;
- repeat structure by NMR and FTIR;
- crystallinity and polymorph by WAXS/DSC; and
- chain-end identity after deliberate phenol-rich or epoxide-rich feeds.

A decisive experiment is to compare the patent's effective 1.02:1 Example-1 feed with a nominal 0.97:1 feed that ignores catalyst-bound BPS. The two should produce different end-group balances and ideal molecular-weight ceilings.

14.5 Phase 4 - thermal and mechanical qualification

14.5.1 Thermal

- DSC and modulated DSC for T_g, melting and recrystallization;
- DMA for storage modulus, tan delta and service-temperature window;
- TGA in nitrogen and air;
- isothermal melt rheology at 230, 250 and 290 C;
- molecular-weight retention after controlled residence times;
- heat-distortion and Vicat tests using current methods; and
- physical-aging and annealing studies near 160 C.

14.5.2 Mechanical

- tensile stress-strain with modulus, yield, break and toughness;
- notched and unnotched impact;
- fracture toughness;
- creep and stress relaxation at elevated temperature;
- fatigue and environmental stress cracking; and
- property retention after water, base, acid and humidity exposure.

A modern study should not treat increasing yield stress after annealing as automatically beneficial. Ductility, impact and fracture behavior must be measured in parallel.

14.6 Phase 5 - application modules

14.6.1 Electronics and dielectric film

Measure:

- film formation and copper adhesion;

- dielectric constant and loss versus frequency and humidity;
- volume/surface resistivity and breakdown strength;
- thermal expansion and warpage in laminate stacks;
- ionic contamination, especially Na, chloride, N and P residues;
- laser-drill/desmeat compatibility; and
- solder-reflow and pressure-cooker aging.

14.6.2 Coating, adhesive and composite matrix

Evaluate:

- lap shear and peel on aluminium, steel, copper, glass and carbon fibre;
- hot-melt open time and melt viscosity;
- crosslinking with representative blocked isocyanate or other OH-reactive chemistry;
- cure conversion and network density;
- phase behavior when blended into epoxy thermosets; and
- interlaminar fracture toughness in composite coupons.

14.6.3 Barrier and membrane

For dense films measure OTR, CO₂ transmission and WVTR under dry and humid conditions. For phase-inversion or electrospun membranes measure pore size, porosity, tensile strength, flux, protein adsorption, fouling, cleanability and sealability. Blend studies with PES should include phase diagrams and leachable fractions.

14.6.4 Bioengineering and bioprocess

Before direct biological contact:

- perform exhaustive and simulated-use extractions;
- quantify BPS, DGE-BPS, oligomers and reaction by-products by LC-MS/MS and high-resolution MS;
- quantify residual solvents and metals;
- test chemical and molecular-weight change after intended sterilization cycles;
- conduct risk-based biological evaluation under the applicable ISO 10993 framework; and
- use worst-case surface-area-to-volume and time/temperature conditions.

14.7 Phase 6 - decision logic

A useful modern material should satisfy a role-specific benchmark, not simply reproduce a 1959 recipe.

- **Free thermoplastic:** stable melt window, reproducible molecular weight, toughness and creep performance competitive with an incumbent.
- **Reactive film/adhesive:** high film strength and adhesion, controlled crosslinking, low ionic and low-molecular-weight residue.
- **Thermoset toughener:** predictable miscibility/phase separation and a measurable fracture-toughness gain without unacceptable T_g loss.
- **Membrane blend:** improved toughness or sealability without flux, fouling or leachables penalty.
- **Bio-contact material:** extractables and biological risk acceptable for the exact exposure route; otherwise restrict use to non-contact structural or device layers.

14.8 Most informative first experiment

A compact first campaign would make the four benchmark architectures at matched absolute molecular weight, then compare T_g/Vicat, water uptake, dielectric behavior, impact and adhesion. Matching molecular weight is crucial: it would distinguish the chemical contribution of the sulfone bridge from the molecular-weight contribution that partially confounds the historical Table 1.

15 Conclusions

15.1 Final judgement

Arie Klootwijk's work was not hiding a non-polymer SO₂ technology. It was more interesting than that interpretation in a different way: it was an early, coherent design for a **reactive high-performance thermoplastic**.

The sulphur dioxide notation points to a rigid neutral diaryl-sulfone bridge that raises thermal and mechanical performance. The equimolar window is a precise chain-length control in bifunctional polymerisation. The solvent requirement is a way to keep growing chains mobile through the last, molecular-weight-defining increments of conversion. The anhydrous catalyst and purity limits protect effective functionality. The final secondary OH groups give adhesion and a latent route to crosslinking.

The strongest modern reading is therefore:

A sulfone-enhanced phenoxy platform designed to be processed as a thermoplastic, with enough built-in hydroxyl chemistry to become an adhesive, reactive film, thermoset modifier or functional interface.

That is genuinely ahead of the ordinary wording about fibres and mouldings. It does not require a hidden cipher; the revolutionary part is the integration of polymer reaction engineering and latent functionality.

15.2 Priority next step

The most informative experimental programme is not to reproduce only one historical example. It is to make four matched-molecular-weight polymers - BPA/DGEBA, BPS/DGEBA, BPA/DGE-BPS and BPS/DGE-BPS - at controlled effective ratios, then compare:

- absolute molecular weight and branching;
- Tg, Vicat, crystallinity and melt stability;
- impact, fracture toughness and creep;
- water uptake and property retention;
- adhesion and controlled post-crosslinking;
- dielectric performance; and
- BPS/DGE-BPS/oligomer extractables.

That programme would separate the sulfone effect from molecular weight, validate the subtle catalyst/ratio bookkeeping, and reveal whether the most valuable modern form is a free thermoplastic, an adhesive film, a composite toughener or a membrane blend component.

A Carothers calculations

For a linear AA + BB system with r defined as the smaller initial functional-group population divided by the larger and p as conversion of the limiting groups:

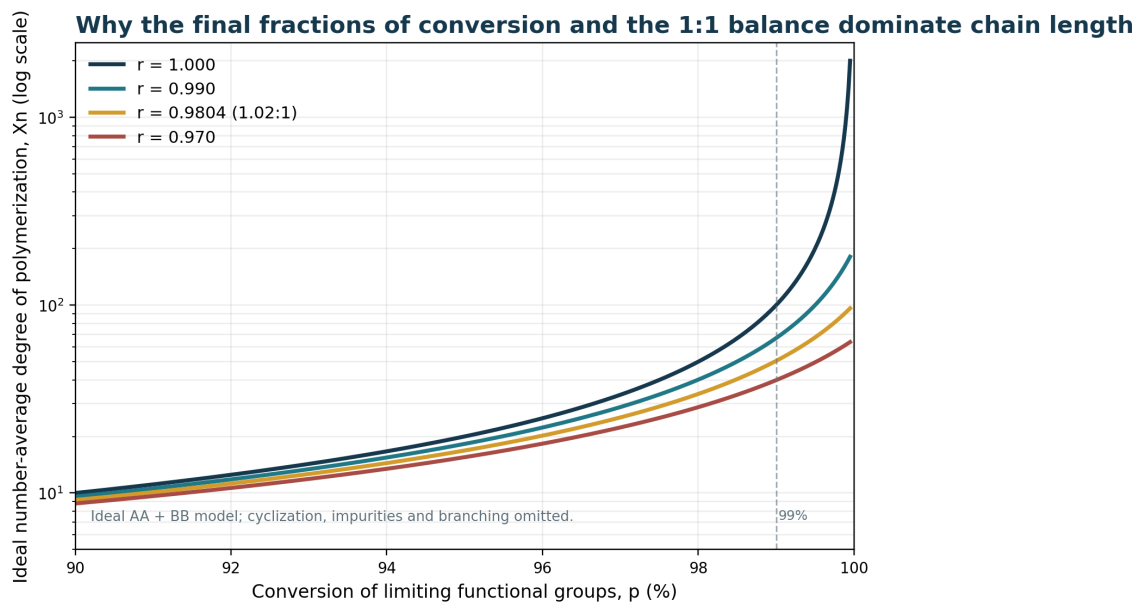
$$X_n = \frac{1 + r}{1 + r - 2rp}$$

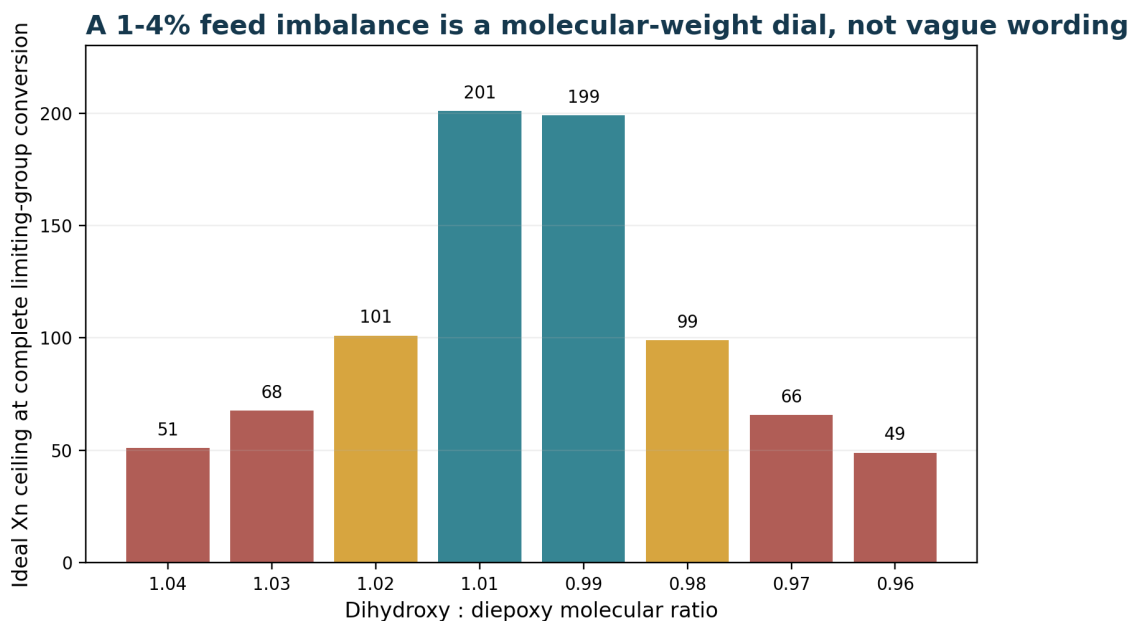
At complete limiting-group conversion:

$$X_{n,\max} = \frac{1 + r}{1 - r}$$

At exact balance, $r = 1$:

$$X_n = \frac{1}{1 - p}$$





Patent preferred range: about 1.02 to 0.99. The exact 1.00 point is omitted because the ideal ceiling diverges.

Nominal dihydroxy:diepoxy ratio	r	Ideal Xn ceiling
1.04:1	0.9615	51
1.02:1	0.9804	101
1.01:1	0.9901	201
1.00:1	1.0000	no finite stoichiometric ceiling in the ideal model
0.99:1	0.9900	199
0.98:1	0.9800	99
0.97:1	0.9700	65.7

These are ideal number-average degrees of polymerisation, not measured values. Cyclisation, unequal reactivity, branching, monofunctional impurities and incomplete conversion reduce them.

B Documentary extracts

B.1 Swedish front matter

The original first page identifies Shell, the inventors and the aryl-SO₂-aryl thermoplastic subject.

SVERIGE  PATENT- OCH REGISTRERINGSVERKET	UTLÄGGNINGSSKRIFT nr 301 717 Int Cl C 08 g 29/04 Kl. 39 c 30 P.ans. nr 5843/60 Inkom den 15 VI 1960 Giltighetsdag den 15 VI 1960 Ans. allmänt tillgänglig den 17 VI 1968 Ans. utlagd och utläggnings- skriften publicerad den 17 VI 1968 Prioritet begärd från den 15 VI 1959 (Nederländerna, 240 201)
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SHELL INTERNATIONALE RESEARCH MAATSCHAPPIJ NV, HAAG, NEDERLÄNDERNA

Uppfinnare: R W Kreps, A Klootwijk och J M Goppel

Ombud: E Lindquist

Sätt att framställa termoplaster av en dihydroxiförening och en diepoxiförening varvid diepoxiföreningen eller båda innehåller en SO₂-grupp

B.2 U.S. comparison table

The U.S. counterpart contains a comparison in which sulfone content is increased across related poly(hydroxy ether) compositions.

3,364

3

TABLE 1

Monomers	Polymer No.				
	1	2	3	4	5
Epoxide:					
Diglycidyl ether of 4,4'-diphenylolpropane	X	X		X	
Diglycidyl ether of 4,4'-diphenylolsulfone			X		
Diglycidyl ether of 3,3', 5,5'-tetramethyl-4,4'-diphenylol-sulfone					X
Phenol:					
4,4'-diphenylolpropane	X				
4,4'-diphenylolsulfone		X	X		
3,3', 5,5'-tetramethyl-4,4'-diphenylolsulfone				X	X
Properties:					
IV, dl./g.	0.57	0.67	0.70	0.55	0.47
Yield Stress, kg./cm. ²	600	750	820	690	880
Tensile impact strength, kg.-cm./cm. ²	265	500	770	360	160
Vicat softening temp., 1 mm. pen., ° C	103	135	175	147	167

In Table 1, Polymer No. 1 is the polymer of bisphenol A and the diglycidyl ether of bisphenol A. In Polymer No. 2 the sulfone is used as the dihydric reactant. In Polymer No. 3 both the diepoxide and dihydric reactant are sulfones. It is seen that yield stress and tensile impact strength are substantially increased as first one and then

C Claims map

Claim	Core limitation	Interpretive significance
1	Dry polar solvent, all species soluble, near-equimolar dihydroxy/diepoxy, aryl-sulfone, IV ≥ 0.4	The invention is the integrated process window.
2	Dihydric phenol geometry	Controls sterics/topology.
3	Di(epoxyalkyl) ether	Defines preferred bifunctional epoxide.
4	BPS + DGE-BPS	Maximum sulfone content in the preferred pair.
5	Product stays dissolved to IV ≥ 0.8	Higher-MW preferred embodiment.
6	Dielectric constant ≥ 10	Higher-polarity solvent preference.
7	Anhydrous catalyst	Protects functionality and conversion.
8	Catalyst is salt of reacting phenol	Integrates catalysis with feed chemistry.
9	Mono-alkali salt	Preferred phenolate form.
10	BPS mp ≥ 240 C	Purity/isomer proxy.
11	DGE-BPS mp ≥ 162 C	Purity proxy, with low chloride preferred in description.

D Data files supplied

The ZIP package includes machine-readable tables:

- 04_Data/patent_examples.csv
- 04_Data/us3364178_table1.csv
- 04_Data/carothers_stoichiometry.csv
- 04_Data/application_assessment.csv
- 04_Data/patent_family.csv
- 04_Data/claims_map.csv

These preserve the distinction between historical units, converted SI values, direct source data and interpretive notes.

E Use limitations

This report is scientific and historical analysis. It is not a legal opinion, a certified translation, a current safety data sheet, a regulatory clearance, or a validated synthesis protocol. Reproduction of the chemistry requires independent laboratory hazard assessment. Biomedical, food-contact and electronics uses require application-specific qualification.

F Full bibliography

Accessed or checked 14 August 2026 unless otherwise noted.

1. **SE 301 717 B**, *Sätt att framställa termoplaster av en dihydroxiförening och en diepoxiförening varvid diepoxiföreningen eller båda innehåller en SO₂-grupp*. Swedish Patent and Registration Office, publication 17 June 1968; Netherlands priority 15 June 1959. Original supplied by the Klootwijk family and bundled in this package.
2. **GB 915,767 A**, Kreps, R. W. F.; Klootwijk, A.; Goppel, J. M., *Improvements in or relating to the production of thermoplastic products derived from di-epoxy compounds*. Published 16 January 1963. <https://patents.google.com/patent/GB915767A/en>
3. **US 3,364,178 A**, Kreps, R. W. F.; Klootwijk, A.; Goppel, J. M., *Thermoplastic poly(hydroxyalkyl diphenyl sulfone) ethers*. Published 16 January 1968. <https://patents.google.com/patent/US3364178A/en>
4. **US 4,108,837 A**, Johnson, R. N.; Farnham, A. G., *Polyarylene polyethers*. Published 22 August 1978. Used here to contrast the fully aromatic SNAr polysulfone/polyethersulfone route and for its independent discussion of stoichiometric and water control. <https://patents.google.com/patent/US4108837A/en>
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7. **US 6,451,878 B1**, Fukuzawa, T.; Hirai, T.; Imura, T.; Ohnuma, Y., *High molecular weight epoxy resin and resinous composition for printed circuit board*. Published 17 September 2002. <https://patents.google.com/patent/US6451878B1/en>
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13. California Office of Environmental Health Hazard Assessment, "Bisphenol S (BPS) Added to the Proposition 65 List for Reproductive Toxicity." The page records female, male and developmental toxicity end-points added in 2023, 2025 and 2025 respectively. <https://oehha.ca.gov/proposition-65/crn/bisphenol-s-bps-added-proposition-65-list-reproductive-toxicity>
14. European Food Safety Authority, "Bisphenol A and other bisphenols" topic page, including EFSA's review activity concerning bisphenol S and the need for further data. <https://www.efsa.europa.eu/en/topics/topic/bisphenol>

F.1 Notes on evidence use

- Patent documents are used as primary historical and technical sources. Their example data are not treated as independently replicated scientific measurements.
- Later patents support technological plausibility and lineage but do not establish that the exact 1959 polymer was commercialized in the later application.

- The 2004 paper confirms the continued material identity “poly(hydroxyether sulfone)” and hydrogen-bonding behavior; its synthetic route is not identical to every claim of SE 301 717 B.
- Current BPS hazard listings concern the low-molecular-weight chemical. They do not by themselves establish the hazard of a purified high-molecular-weight polymer, but they make residual monomer, oligomer and degradation-product analysis essential.

G Source and method notes

G.1 Source hierarchy

1. User-supplied Swedish patent facsimile, SE 301 717 B.
2. Clean English translation prepared from the Swedish document, cross-checked against GB 915,767 A and DE 1,190,672 B.
3. U.S. counterpart US 3,364,178 A, which contains product-focused language and a comparison table not reproduced in exactly the same form in the Swedish publication.
4. Later patent and journal records used only where explicitly labelled as later context or extrapolation.
5. Polymer-chemistry equations and calculations used to interpret the patent's ratio limits.

G.2 Calculations

- Historical yield stress was converted using $1 \text{ kgf/cm}^2 = 0.0980665 \text{ MPa}$.
- The Carothers-Flory imbalance equation used is $X_n = (1+r)/(1+r-2rp)$, with r defined as the smaller functional-group population divided by the larger.
- At full limiting-group conversion, $X_{n, \text{max}} = (1+r)/(1-r)$.
- Example 1 effective dihydroxy feed is interpreted as $0.97 \text{ mol neutral BPS} + 0.05 \text{ mol monosodium BPS} = 1.02 \text{ mol}$, because the patent expressly says phenol present as salt is included in the ratio.
- Example 4's 2.5 g BPS stopping charge is approximately 0.010 mol using a BPS molar mass near 250.27 g/mol.
- Example 4 intrinsic-viscosity change from 1.37 to 1.29 is approximately a 5.8% decrease; this is not converted into molecular-weight loss because the exponent and calibration are unknown.

G.3 Terminology

- BPS: bisphenol S, 4,4'-sulfonyldiphenol, historically p,p'-diphenylol sulfone.
- DGE-BPS: diglycidyl ether of bisphenol S.
- DGEBA: diglycidyl ether of bisphenol A.
- poly(hydroxy ether): linear polymer formed by repeated phenol/epoxide ring-opening, with a secondary hydroxyl generated at each event; often associated with phenoxy-resin terminology.
- PSU/PESU: commercial fully aromatic polysulfone/polyethersulfone families typically prepared by nucleophilic aromatic substitution. These are structurally distinct from the Klootwijk poly(hydroxy ether).

G.4 Deliberate limits

This review does not claim:

- secret or coded intent;
- direct commercial continuity between every later cited patent and SE 301 717 B;
- current freedom to operate, patent validity or ownership;
- suitability for food, medical or implant use;
- that BPS monomer hazard is identical to polymer hazard; or
- that historical tests meet present qualification standards.

G.5 Privacy

The package identifies the recipient only as Tom Klootwijk. Personal registry numbers, dates of birth and genealogy URLs supplied in conversation were not reproduced.